TECHNIQUE MASTERING EXERCISES

E - ALGEBRAIC OPERATIONS WITH EXPONENTIALS, LOGARITHMS, ETC

1.

a. $2^{\frac{1}{2}}$ **b.** $2^{\frac{3}{4}}$ **c.** 2^{1+4x}

d. 2^{2y} **e.** 2^{x^2} **f.** 2^6

2.

a. $6a^7b$ **b.** $3\ x^{n+4}$ **c.** $\frac{1}{81x^8}y^{12}$ **d.** $3^{(x+1)^2}$

e. $x^{\frac{1-y}{y}}$ **f.** $3|x|$ **g.** $\frac{a}{2b^3}$ **h.** $\frac{1}{(\sqrt{x})^3}$

i. $\frac{27}{14}\sqrt[4]{x}\,y$ **j.** 2^{-4n} **k.** $\frac{27}{8a^3}$ **l.** $x^{7n}y^n$

m. $3^{n+2}2^{3n-4}$ **n.** $\sqrt[6]{a^5}$ **o.** $\frac{1}{3}$ **p.** $\frac{1}{4}$

q. $\frac{2}{25}$ **t.** $a-2\sqrt{ab^3}+b^3$ **s.** $\frac{x}{a-x}$ **t.** $\frac{1}{9}$

3.

a. $\log x+\log y-\log z$ **b.** $\frac{1}{2}\log x-\frac{1}{2}\log y$ **c.** $2\log x+3\log y$

d. $-2\log x-\log y$ **e.** $\ln x+2\ln y-\ln(x+y)$ **f.** $\frac{1}{3}\ln(x^2+1)$

g. $\frac{2}{3}\ln x+\frac{1}{3}\ln y$ **h.** $\frac{3}{2}\ln x-\frac{1}{2}\ln y$ **i.** $\log x-\log z-\log y$

j. $3\ln(x+y)-\ln x-2\ln y$

4.

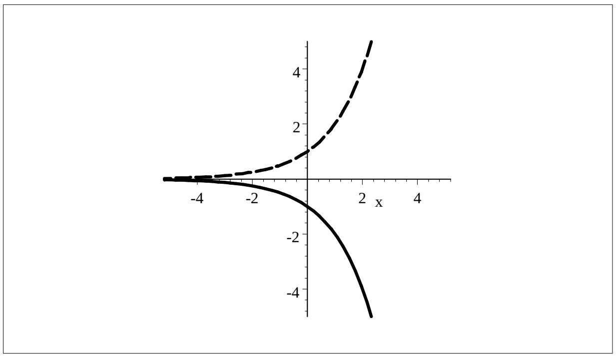
a. $\ln x^3$ **b.** $\ln\frac{x^3}{\sqrt{y}}$ **c.** $\ln\sqrt{\frac{x}{y^3}}$ **d.** $\ln[x^2(x+y)]$

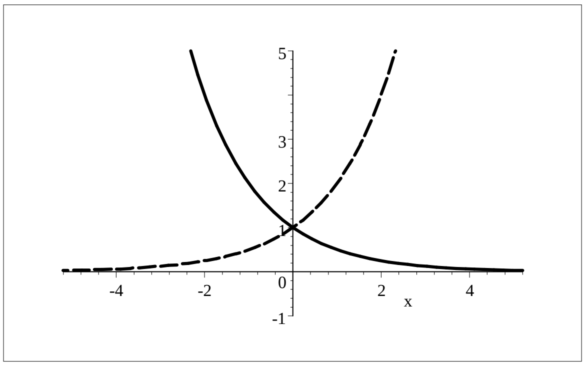
e. $\ln\frac{x^2z}{y^3}$ **f.** $\ln\frac{y(y-2)}{x^2}$ **g.** $\ln\sqrt{\frac{xz^{15}}{y^5}}$ **h.** $\log\sqrt{\frac{x}{y^5z^3}}$

i. $\ln\frac{5x^3}{16y}$ **j.** $\log(xy^2)$

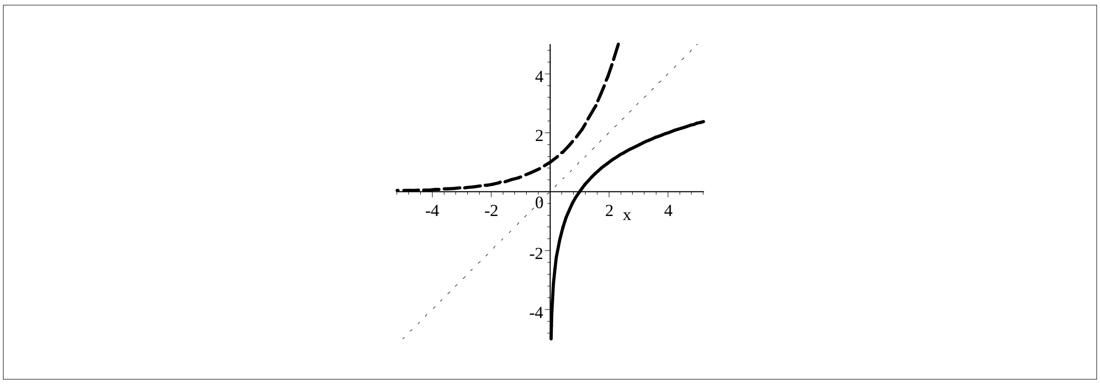
5. **a.** $3y$ **b.** $\frac{1}{3}(x+y)$ **c.** $x+\frac{y}{2}$

6. **a.** $g(x) = -2^x$ **b.** $h(x) = 2^{-x}$ **c.** $k(x) = \log_2 x$





$\left[g \ \underline{\quad\quad},\ f\ \ ---\right]$ $\left[h \ \underline{\quad\quad},\ f\ \ ---\right]$



$\left[k \ \underline{\quad\quad},\ f\ \ ---,\ y = x\ \ \ldots\ldots\right]$

7. $\log_b a = \frac{1}{3}$

8.

$$y = \frac{r}{1 + ce^{-at}} \Rightarrow e^{-at} = \frac{r-y}{cy}$$

$$\Rightarrow -at = \ln\left(\frac{r-y}{cy}\right)$$

9. $f(\frac{1}{b}) = -1$ and $f(\sqrt{b}) = \frac{1}{2}$

10. $b = 100$

11.

a. $4e^{4x} + x^{4e}$ **b.** $2e^{\frac{2x^2+3}{x}}, x > 0$ **c.** $2e^{2x}$ **d.** $e^{-\frac{x}{2}}$

e. $3x$ **f.** $7x,\ x > 0$ **g.** $\ln 90 + 3x, x \in \mathbb{R}$ **h.** $\frac{x^2}{y},\ \frac{x^2}{y} > 0$ and $y \neq 0$.

i. $2\ln x, x > 0$ **j.** $\frac{2x+1}{x}, x > 0$ and $x \neq 0$

12. **a.** $x = 3$ **b.** $x = \frac{1}{4}\ln 12$ **c.** $x = \frac{1}{4}e^{12}$

d. $x = 9$ **e.** $x = \frac{8}{9}$ **f.** $x = \frac{1}{4}e^2$